

Remainder Theorem & Factor Theorem

Algebra Worksheet · Grade 9–11

Name: _____

Date: _____

Learning Objectives

- Use the Remainder Theorem to evaluate a polynomial function at a given value using synthetic division
- Apply the Factor Theorem to determine whether a given binomial is a factor of a polynomial
- Find all rational roots of a polynomial using the Rational Root Theorem and synthetic division

Problems

1. Use the Remainder Theorem and synthetic division to find $f(2)$:

$$f(x) = x^3 - 4x^2 + 5x + 3$$

2. Use the Remainder Theorem and synthetic division to find $f(2)$:

$$f(x) = -x^3 + 6x - 7$$

3. Use the Remainder Theorem to find $f(-3)$:

$$f(x) = 2x^3 + x^2 - 5x + 4$$

4. Use the Remainder Theorem to find $f(4)$:

$$f(x) = x^3 - 6x^2 + 3x - 2$$

5. Use the Factor Theorem to determine whether $(x + 1)$ is a factor of the polynomial below. Show your work using synthetic division.

$$f(x) = x^3 + 9x^2 + 23x + 15$$

6. Use the Factor Theorem to determine whether $(x - 3)$ is a factor of the polynomial below.



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$$f(x) = x^3 - 5x^2 + 2x + 8$$

7. List all possible rational roots of the polynomial below using the Rational Root Theorem, then find one actual root using synthetic division.

$$f(x) = x^3 + 2x^2 - 5x - 6$$

8. Find all real roots of the polynomial below using the Rational Root Theorem and synthetic division.

$$f(x) = x^3 + 9x^2 + 23x + 15$$

9. Find all real roots of the polynomial below using the Rational Root Theorem and synthetic division.

$$f(x) = 2x^3 - 3x^2 - 11x + 6$$

10. A polynomial $f(x)$ of degree 3 has leading coefficient 1 and constant term -12. Using the Rational Root Theorem and synthetic division, find all real roots of the polynomial below and write it in fully factored form.

$$f(x) = x^3 - x^2 - 8x + 12$$



Remainder Theorem & Factor Theorem — Answer Key

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Answer Key

1. Answer: $f(2) = 5$

- Set up synthetic division with divisor 2 and coefficients 1, -4, 5, 3
- Bring down 1; multiply $1 \times 2 = 2$, add to -4 to get -2
- Multiply $-2 \times 2 = -4$, add to 5 to get 1
- Multiply $1 \times 2 = 2$, add to 3 to get remainder 5
- By the Remainder Theorem, $f(2) = 5$

2. Answer: $f(2) = -3$

- Rewrite with all terms: $-x^3 + 0x^2 + 6x - 7$, so coefficients are -1, 0, 6, -7
- Set up synthetic division with divisor 2
- Bring down -1; multiply $-1 \times 2 = -2$, add to 0 to get -2
- Multiply $-2 \times 2 = -4$, add to 6 to get 2
- Multiply $2 \times 2 = 4$, add to -7 to get remainder -3
- By the Remainder Theorem, $f(2) = -3$

3. Answer: $f(-3) = -26$

- Coefficients are 2, 1, -5, 4; divisor is -3
- Bring down 2; multiply $2 \times (-3) = -6$, add to 1 to get -5
- Multiply $-5 \times (-3) = 15$, add to -5 to get 10
- Multiply $10 \times (-3) = -30$, add to 4 to get remainder -26
- By the Remainder Theorem, $f(-3) = -26$

4. Answer: $f(4) = -18$

- Coefficients are 1, -6, 3, -2; divisor is 4
- Bring down 1; multiply $1 \times 4 = 4$, add to -6 to get -2
- Multiply $-2 \times 4 = -8$, add to 3 to get -5
- Multiply $-5 \times 4 = -20$, add to -2 to get remainder -22
- By the Remainder Theorem, $f(4) = -22$

5. Answer: Yes, $(x + 1)$ is a factor because $f(-1) = 0$

- If $(x + 1)$ is a factor, then $f(-1)$ must equal 0
- Coefficients are 1, 9, 23, 15; divisor is -1
- Bring down 1; multiply $1 \times (-1) = -1$, add to 9 to get 8
- Multiply $8 \times (-1) = -8$, add to 23 to get 15
- Multiply $15 \times (-1) = -15$, add to 15 to get remainder 0
- Remainder = 0, so $(x + 1)$ is a factor



6. Answer: No, $(x - 3)$ is NOT a factor because $f(3) = -4$

- If $(x - 3)$ is a factor, then $f(3)$ must equal 0
- Coefficients are 1, -5, 2, 8; divisor is 3
- Bring down 1; multiply $1 \times 3 = 3$, add to -5 to get -2
- Multiply $-2 \times 3 = -6$, add to 2 to get -4
- Multiply $-4 \times 3 = -12$, add to 8 to get remainder -4
- Remainder $\neq 0$, so $(x - 3)$ is NOT a factor

7. Answer: Possible roots: $\pm 1, \pm 2, \pm 3, \pm 6$; one root is $x = 2$

- Factors of leading coefficient (1): ± 1
- Factors of constant (-6): $\pm 1, \pm 2, \pm 3, \pm 6$
- Possible rational roots: $\pm 1, \pm 2, \pm 3, \pm 6$
- Test $x = 2$ with coefficients 1, 2, -5, -6
- Bring down 1; $1 \times 2 = 2$, add to 2 = 4; $4 \times 2 = 8$, add to -5 = 3; $3 \times 2 = 6$, add to -6 = 0
- Remainder = 0, so $x = 2$ is a root

8. Answer: $x = -1, x = -3, x = -5$

- Possible rational roots from factors of 15 over factors of 1: $\pm 1, \pm 3, \pm 5, \pm 15$
- Test $x = -1$: remainder = 0, so $(x + 1)$ is a factor; quotient is $x^2 + 8x + 15$
- Factor $x^2 + 8x + 15 = (x + 3)(x + 5)$
- Set each factor to zero: $x = -1, x = -3, x = -5$

9. Answer: $x = 3, x = -2, x = 1/2$

- Possible rational roots: $\pm 1, \pm 2, \pm 3, \pm 6, \pm 1/2, \pm 3/2$
- Test $x = 3$ with coefficients 2, -3, -11, 6: remainder = 0; quotient is $2x^2 + 3x - 2$
- Factor $2x^2 + 3x - 2 = (2x - 1)(x + 2)$
- Set each factor to zero: $x = 3, x = 1/2, x = -2$

10. Answer: $x = 2$ (double root), $x = -3$; factored form: $(x - 2)^2(x + 3)$

- Possible rational roots from factors of 12: $\pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 12$
- Test $x = 2$ with coefficients 1, -1, -8, 12: remainder = 0; quotient is $x^2 + x - 6$
- Factor $x^2 + x - 6 = (x - 2)(x + 3)$
- So $f(x) = (x - 2)(x - 2)(x + 3) = (x - 2)^2(x + 3)$
- Roots are $x = 2$ (multiplicity 2) and $x = -3$

